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Using Cross-Domain Data to Predict Syllogistic Reasoning Behavior

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Motivation

- The ability to reason and draw inferences is central to human cognition and therefore an important field of research
- Across various domains of reasoning, research identified various strategies, mechanisms and effects, which have been refined into theories [1]
- Cognitive models then instantiate the theories, making the ideas testable
- However, modeling endeavors suffer from several problems:

- Models and theories are incomplete → Some mechanisms and effects are not considered
- Models are often not implemented and built with extensibility in mind
- Findings are hard to transfer between models → No unified structure
- Models are often not made with individual reasoners in mind

Syllogistic Reasoning & Dataset

- Syllogistic reasoning is one of the oldest domains in human reasoning research
- Well-defined domain with a plethora of theories and models [2]

All **samurai** are **warriors**.
Some **warriors** are **ninjas**.
What, if anything, follows?

- Two quantifier statements connected via a **middle term**
- Task: Conclude what holds for the other terms (**samurai**, **ninjas**)

- We used data from a study 95 participants [3]
 - Participants solved all **64 traditional syllogisms**
 - Multiple-Choice responses or „No valid conclusion“ (NVC)

Task Processing	Response Candidate Generation				Response Candidate Alteration			
Change Interpretation	Derive NVC directly		Gen. Quantifiers		Extend Conclusions		Test Conclusions	
Conversion	NVC Rules*		Quantifier		Extensions*		Conclusion Tests*	
Path 8.0%	Two-Some 19.3%	Atmosphere 48.6%	Figure Effect 0.9%	p-Entailment 30.3%	O-Heuristic 24.5%	Grice 31.3%	Preferences*	
Illicit -	Two-Negation 22.7%	Matching 28.2%	MMT 0.5%	Grice Imp. 48.1%	MMT 11.1%	Matching 7.8%		
Transitive 6.6%	PartNeg 15.5%	MMT 4.5%	PHM 23.4%		Max-Heur. A 2.3%	PHM Inform. 15.0%		
	FirstNeg 15.5%	PHM 18.7%	Bidirectional 75.2%		Max-Heur. I 5.3%	Figure Effect 30.3%		
	Path 28.9%				Max-Heur. E 7.3%	Atmosphere 2.5%		

Method

Core Idea: Automatically generate a process model to capture an individual reasoner based on mechanisms, processes and effects identified from research.

- Requires a generalized structure
- Every cognitive process consists of steps
 - Represent all mechanisms and effects as **actions**
 - Find a structure that fits for the majority of the state-of-the-art

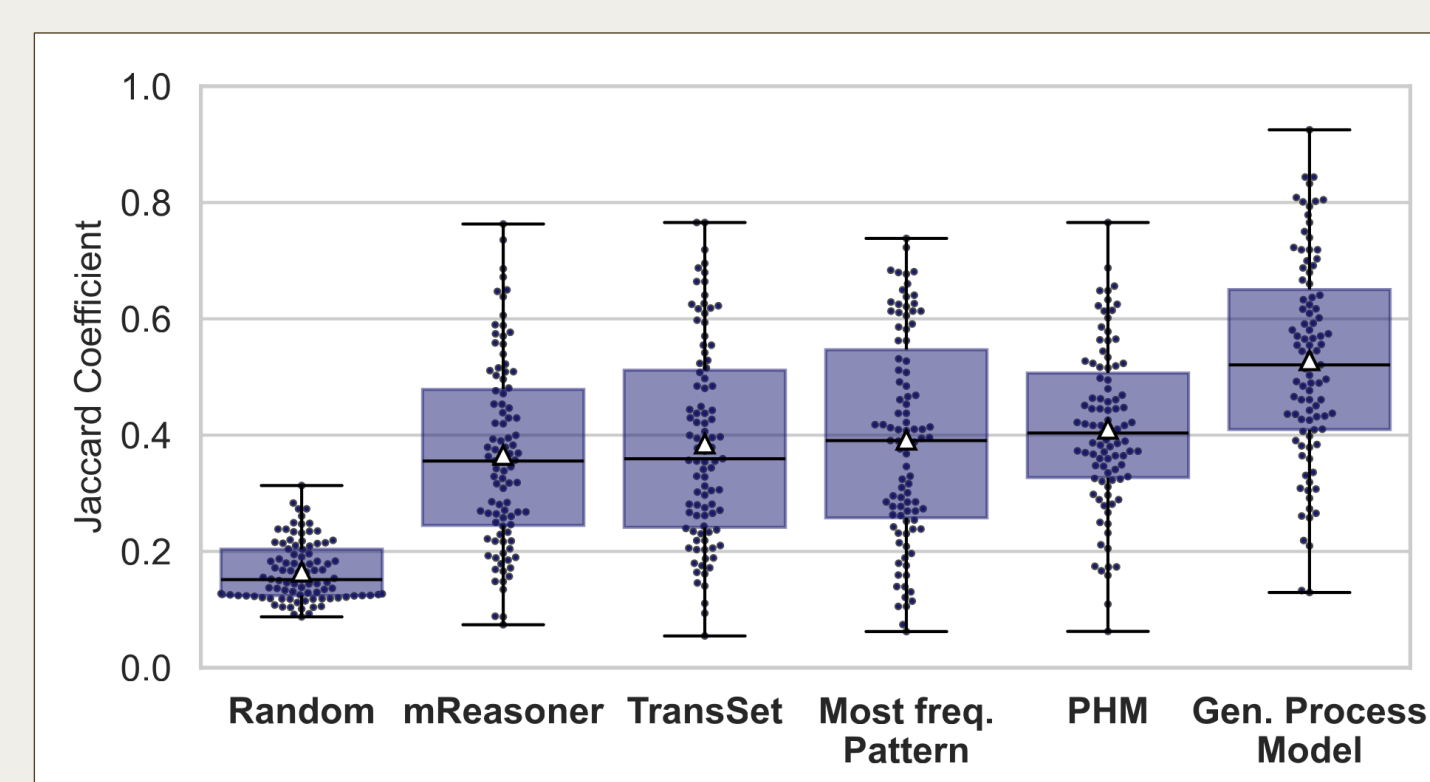
Three types of actions:

1. **Task Processing:** interpreting the task and its meaning
2. **Response Candidate Generation:** builds the initial mental representations and solution candidates
3. **Response Candidate Alteration:** Checks, extends, alters or discards the response candidates if necessary

- We then implemented mechanisms, effects and processes identified in the state-of-the-art for syllogistic reasoning (e.g. [4,5,6])
- We introduced phases within each type of actions to refine the process for the domain of syllogistic reasoning
 - Now, an **individualized process model** can be generated for each participant by searching the best combination of actions

Results

- Our method can account for **78.8%** of all observed responses (when considering them as independent task-response pairs)
- When accounting for an individual reasoner (i.e., all 64 tasks of a person), state-of-the-art models were outperformed substantially



- We used leave-one-out-CV to evaluate predictive performance
- Jaccard coefficient was chosen as metric to compare sets of responses [7]
- Most-freq. pattern is the upper bound when not considering inter-individual differences

Conclusions

- Our method **generates process models automatically** for individual reasoners while preserving their theoretical foundation
- The process models achieve **high predictive accuracy** and outperform current state-of-the-art models in the domain of syllogistic reasoning
- Our method can serve as a **framework** to streamline modeling endeavors and facilitate transfer between theories and models, serving as a step towards unifying computational models of human reasoning

References

- [1] Manktelow, K. I. (1999). Reasoning and thinking. Psychology Press.
- [2] Khemlani, S. S., & Johnson-Laird, P. N. (2012). Theories of the syllogism: A meta-analysis. *Psychological Bulletin*, 138(3), 427–457.
- [3] Brand, D., & Ragni, M. (2025). Using cross-domain data to predict syllogistic reasoning behavior. In D. Barner, N. R. Bramley, A. Ruggeri, & C. M. Walker (Eds.), *Proceedings of the 47th Annual Meeting of the Cognitive Science Society* (pp. 4370–4377).
- [4] Brand, D., Riesterer, N., & Ragni, M. (2020). ExtendingTransSet: An individualized model for human syllogistic reasoning. In T. C. Stewart (Ed.), *Proceedings of the 18th International Conference on Cognitive Modeling* (pp. 17–22).
- [5] Chater, N., & Oaksford, M. (1999). The probability heuristics model of syllogistic reasoning. *Cognitive Psychology*, 38(2), 191–258
- [6] Khemlani, S. S., & Johnson-Laird, P. N. (2013). The processes of inference. *Argument & Computation*, 4(1), 4–20.
- [7] Brand, D., & Ragni, M. (2023). Effect of response format on syllogistic reasoning. In M. Goldwater, F. K. Anggoro, B. K. Hayes, & D. C. Ong (Eds.), *Proceedings of the 45th Annual Conference of the Cognitive Science Society* (pp. 2408–2414).



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